

Create a Flex database

Shows how to create a Flex database on Redis Cloud and describes the best use cases for Flex.

Redis Cloud

This is a snapshot of the Redis Docs from July 13, 2026.

See <https://redis.io/docs/latest/operate/rc/databases/create-database/create-flex-database/> for the current version.

Flex allows your dataset to span both high-performance RAM and cost-efficient dedicated Flash memory. Flex automatically manages data placement between the two tiers, keeping frequently accessed (“hot”) data in RAM for sub-millisecond latency, while moving less active (“warm”) data to Flash to optimize capacity and cost. This dual memory architecture delivers predictable performance at scale, enabling larger datasets without compromising speed or operational simplicity.

Flex databases are currently compatible with most existing Redis applications, except for applications that use Search and Query and Time Series.

Flex is available on both Redis Cloud Essentials and Redis Cloud Pro.

When to use Flex

Flex is ideal for workloads that demand large-scale, low-latency data access with the flexibility to optimize cost and performance.

Consider Flex when you need to:

- Run Redis at terabyte scale while maintaining high throughput and sub-10 ms latency.

- Power real-time feature stores for machine learning applications such as fraud detection, recommendation systems, and personalization engines.
- Operate large distributed caches that require elastic scaling and consistent performance under heavy load.
- Optimize infrastructure cost by combining high-speed RAM with cost-efficient Flash storage through Flex's automatic data tiering.

Flex is **not** a durable data store. It is designed for performance, elasticity, and scalability, not for long-term data persistence. While Flex can temporarily retain data in memory or Flash, it should not be used as a primary system of record or persistent storage layer.

For workloads that require durability and recovery across restarts or failures, use Redis Cloud's [Data persistence](#) features.

Create a Flex database

Redis Cloud Essentials

Before creating a Redis Cloud database, you need to [create an account](#).

To create a database in your Redis Cloud account:

1. Sign in to the [Redis Cloud console](#).
2. Select the **New database** button.

A blue rectangular button with rounded corners containing the text "New database" in white.

This displays the **Create database** screen.

3. Select the type of [subscription](#) you need. For this guide, select **Essentials**.

Choose your plan

Essentials
Shared infrastructure optimized for smaller workloads and cost-efficient scale

From
\$0.007 /hour
\$5 /month

[Start with Essentials](#)

 Scale anytime

Pro
Dedicated infrastructure for larger, mission-critical workloads with greater control and isolation

From
\$0.014 /hour
Minimum \$200 /month

[Get Pro](#)

Shared deployment
Shared infrastructure designed to scale with you

Basic security
SAML SSO, RBAC, encryption in transit and at rest

Single DB, RAM + SSD (Flex)
Auto-tiering from 250 MB to 100 GB across RAM and Flash

Availability and backup
Up to 99.99% uptime with multi-AZ high availability and backups

Dedicated deployment
Dedicated resources with full network isolation

Advanced security
VPC, private networking, and Active-Active geo

Multiple DBs, RAM + SSD (Flex)
Auto-tiering with no upper bound across RAM and Flash

Availability and backup
Up to 99.999% uptime with enhanced data durability

Not ready? [Try 30 MB for free](#) - Designed for testing and learning

Want better Pro prices? [Check our annual plan](#)

After you select Essentials, the rest of the database details will appear. Select Flex (RAM + SSD) to use Flex.

Select cloud configuration

Cloud vendor

aws Amazon Web Services

Region

US East (N. Virginia) us-east-1

Name

database-MQ19M67H

Database version

8.4

Type

Redis

- Redis will generate a database name for you. If you want to change it, you can do so in the **Database name** field.
- Choose a **Region** on Amazon Web Services for your database. See [Supported regions](#) for a list of supported regions by cloud vendor.
- Select the **Database version** you want to use.
- Select your desired memory limit.

Plan details

Total size & plan

250 MB 1 GB 2.5 GB 5 GB 12 GB 25 GB 50 GB 100 GB

RAM

RAM+SSD (Flex)

1.25 GB Dataset

1.25 GB Replica

500 Ops/sec

\$15/mo Price

These values scale with total size

✓ 100GB Network usage

✓ 8 CIDR allow rules

✓ 2500 Connections max

✓ Daily and instant backups

✓ Standard support

Protect your data Recommended

High availability (replication)

Single zone - up to 99.9% uptime

Database replication spanning multiple nodes with automatic failover

Data persistence

Append-only file every 1 sec

Allows for data recovery in the event of memory loss or other catastrophic failure.

For a comparison of available plans, see [Redis Cloud Essentials plans](#).

All Flex plans on Redis Cloud Essentials have a default RAM percentage of 10%.

8. Choose your **High availability (replication)** settings from the list.

Redis Cloud supports the following high availability settings with Flex:

- **None:** You will have a single copy of your database without replication.
- **Single-Zone:** Your database will have a primary and a replica located in the same cloud zone. If anything happens to the primary, the replica takes over and becomes the new primary.

See [High availability](#) for more information about these settings.

9. Choose your **Data persistence** settings from the list.

Redis Cloud supports the following data persistence options:

- An **Append-Only File** maintains a record (sometimes called a *redo log* or *journal*) of write operations. This allows the data to be restored by using the record to reconstruct the database up to the point of failure. For Essentials databases, Redis updates the Append-Only file every second.
- A **Snapshot** is a copy of the in-memory database, taken at periodic intervals (one, six, or twelve hours). You can restore data to the snapshot's point in time.

See [Data persistence](#) for more information about these settings.

10. Select the **Database version** you want to use.

11. Enter your payment details.

If you haven't previously entered a payment method, use the **Add Credit Card** button to add one.



If you have not already added a business address to your account, you must enter one. Redis uses your business address for communication, invoicing, and tax purposes.

Add credit card

Credit card details

STREET ADDRESS

1234 1234 1234 1234



MM / YY

CVC

123

First name

Last name

By selecting "Add credit card", I acknowledge my payment details will be saved for recurring and future purchases. You may see a temporary charge as part of the card authorization process.

Business address

Street address 1

Street address 2 (Optional)

Country

City

Zip code

VAT / GST ID (Optional) 

This information will be used by Redis for communication, invoicing, and tax calculation, where applicable. See our [Privacy Policy](#).

Billing address ☐ Same as business address

Street address 1

Street address 2 (Optional)

Country

City

Zip code

If you already have a business address, you don't need to enter one here. See [Account settings](#) to learn how to update your business address.

Add credit card

Credit card details

Previously used credit cards 

1234 1234 1234 1234



MM / YY

CVC 

First name

Last name

By selecting "Add credit card", I acknowledge my payment details will be saved for recurring and future purchases. You may see a temporary charge as part of the card authorization process.

Billing address ☐ Same as business address

Street address 1

Street address 2 (Optional)

Country 

Select... 

City

Zip code

If the card's billing address is the same as your account's business address, select **Same as business address** to fill the billing address with your business address details.

12. Select **Confirm & pay** to create your database.

Confirm & pay

When you create your database, there's a brief pause while your request is processed and then the **Database details** page appears.

Redis Cloud Pro

To create a Flex database on Redis Cloud Pro, [create a new Pro database with custom settings](#).

In the **Advanced options** of the **Setup** tab, select **Redis Flex**.

Redis Flex

Reduce the costs of large datasets by supplementing main memory with high-performance SSDs.

[Learn more ↗](#)



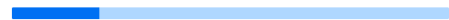
During the Sizing step, when you are provisioning your databases, you can select the RAM percentage for your database. The default is 20%, but you can select a percentage between 10% and 50%. Lower RAM percentages reduce cost but may increase latency, while higher RAM percentages improve throughput and latency at higher cost. See [Choosing the right RAM ratio](#) to choose the right RAM percentage for your database.

RAM Percentage ⓘ

The percentage of dataset to allocate to RAM

• RAM 20% (50GB)

• SSD 80% (200GB)



Continue with the instructions to [create your database](#).

Troubleshooting

See [Troubleshooting Redis Flex performance questions](#) on the Redis Support knowledge base to learn how to troubleshoot Redis Flex databases.

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